# GRISHMA BALGI

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**Test, Validation & Sensor Instrumentation**

## EDUCATION

**New York University | May 2027**
M.S. Mechatronics and Robotics | GPA: 3.834/4.0

**Veermata Jijabai Technological Institute | June 2025**
B.Tech. Mechanical Engineering; Honors in Automobile Engineering | GPA: 9.04/10

## TECHNICAL SKILLS

**Instrumentation:** HX711 load-cell acquisition, vibration/acoustic sensing, serial data logging

**Analysis:** Python, NumPy, Pandas, Matplotlib, MATLAB

**Systems:** Arduino, STM32, Dynamixel, electromechanical integration, CAD fixtures

## EXPERIENCE

**Prema Lab, New York University | Sep. 2025 - Present**
Graduate Research Assistant | Robotics fabrication, instrumentation and simulation

- Fabricated a robotic-fin test rig and automated Silhouette Portrait 4 cutting through Python USB commands, generating coordinate-based paths for candidate fin geometries.
- Developed a motor-control interface and Arduino acquisition firmware to vary oscillation waveform, amplitude and frequency; logged load-cell readings alongside commanded and measured motor positions.

**Schneider Electric | June 2024 - June 2025**
Engineering Intern | Mechanical modelling and sensor systems

- Worked on reduced-order modelling of mechanical systems using Creo and Altair tools, preparing simulation-derived data and exploring models for circuit-breaker mechanisms.
- Built experience in sensor selection, vibration data acquisition and Python-based analysis for circuit-breaker condition monitoring.

## SELECTED PROJECTS

**Multi-Sensor Circuit-Breaker Fault Detection**
Python, Arduino, scikit-learn, sensor acquisition | Undergraduate team project | 2024-2025

- Developed a circuit-breaker monitoring prototype using vibration and acoustic sensing, sensor-data preprocessing and classification of spring-removal conditions.
- Compared classical classifiers and developed a CSV-based prediction workflow; kept synthetic-data experiments distinct from physical measurements and deployment claims.

**VJTI Racing | Competition Electric ATV**
SolidWorks, Creo, Ansys, fabrication | Student engineering team | 2022-2024

- Contributed to mechanical subsystem design, structural analysis and fabrication for a competition electric ATV, including suspension, drivetrain mounting and chassis components.
- Supported wiring-harness and motor-controller integration, prototype fixtures, vehicle testing and fault diagnosis across iterative builds.